

Lecture #14: Change of Variables

MATH 321 – Mathematical Statistics

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Finding the Distribution of a Function of Random Variables

We've talked a bit about how to find the distribution of a function of random variables, for certain classes of functions, but let's focus some more on it.

There are a few ways we can try to find the distribution of a function of a random variable (or random variables) that we know the distribution of:

- ① The Method of Moment-Generating Functions
- ② The Method of Distribution Functions
- ③ The Method of Transformations
- ④ Using the Jacobian

The Method of Moment-Generating Functions

There are two important properties of the MGF to remember:

- ① If $Y = aX$ where $a \in \mathbb{R}$ is a constant, then:

$$M_Y(t) = M_X(at).$$

- ② If $Z = \sum_{i=1}^n X_i$, then

$$M_Z(t) = \prod_{i=1}^n M_{X_i}(t).$$

The Method of Moment-Generating Functions

Summary of the Moment-Generating Function Method

Let U be a function of the random variables $Y_1, Y_2, \dots, Y_n$.

- 1 Find the moment-generating function for U , $m_U(t)$.
- 2 Compare $m_U(t)$ with other well-known moment-generating functions. If $m_U(t) = m_V(t)$ for all values of t , this implies that U and V have identical distributions.

Example: The Method of Moment-Generating Functions

Example: Suppose that $X_i \stackrel{i.i.d.}{\sim} \text{Exponential}(\beta)$. If $Y = \sum_{i=1}^n X_i$, show that the distribution of Y is:

$$Y \sim \text{Gamma}(\alpha = n, \beta = \beta).$$

Example: Let Z be a normally distributed random variable with mean $\mu = 0$ and variance $\sigma^2 = 1$. Use the method of moment-generating functions to find the probability distribution of Z^2 .

The Method of Distribution Functions

Example: A process for refining sugar yields up to 1 ton of pure sugar per day, but the actual amount produced, Y , is a random variable because of machine breakdowns and other slowdowns. Suppose that Y has density function given by

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The company is paid at the rate of \$300 per ton for the refined sugar, but it also has a fixed overhead cost of \$100 per day. Thus the daily profit, in hundreds of dollars, is $U = 3Y - 1$. Find the probability density function for U .

The Method of Distribution Functions

Summary of the Distribution Function Method

Let U be a function of the random variables $Y_1, Y_2, \dots, Y_n$.

- 1 Find the region $U = u$ in the $(y_1, y_2, \dots, y_n)$ space.
- 2 Find the region $U \leq u$.
- 3 Find $F_U(u) = P(U \leq u)$ by integrating $f(y_1, y_2, \dots, y_n)$ over the region $U \leq u$.
- 4 Find the density function $f_U(u)$ by differentiating $F_U(u)$. Thus,

$$f_U(u) = \frac{dF_U(u)}{du}.$$

Example: The Method of Distribution Functions

Example: Let Y have probability density function given by

$$f_Y(y) = \begin{cases} \frac{y+1}{2}, & -1 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example: Let U be a uniform random variable on the interval $(0, 1)$. Find a transformation $G(U)$ such that $G(U)$ possesses an exponential distribution with mean β .

The Method of Transformations

We can take the ideas in that last example further.

Summary of the Transformation Method

Let $U = h(Y)$, where $h(y)$ is either an increasing or decreasing function of y for all y such that $f_Y(y) > 0$.

① Find the inverse function, $y = h^{-1}(u)$.

② Evaluate $\frac{dh^{-1}}{du} = \frac{d[h^{-1}(u)]}{du}$.

③ Find $f_U(u)$ by

$$f_U(u) = f_Y[h^{-1}(u)] \left| \frac{dh^{-1}}{du} \right|.$$

Example: The Method of Transformations

Example: Let Y have the probability density function given by:

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the density function of $U = -4Y + 3$.

Example: The Method of Transformations

Example: Consider a random variable Y_1 , the proportion of impurities in a chemical sample, and Y_2 , the proportion of type I impurities among all impurities in the sample. The joint density function was given by

$$f(y_1, y_2) = \begin{cases} 2(1 - y_1), & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

We are interested in $U = Y_1 Y_2$, which is the proportion of type I impurities in the sample. Find the probability density function for U .

Multivariable Transformations Using Jacobians

The Bivariate Transformation Method

Suppose that Y_1 and Y_2 are continuous random variables with joint density function $f_{Y_1, Y_2}(y_1, y_2)$ and that for all (y_1, y_2) , such that $f_{Y_1, Y_2}(y_1, y_2) > 0$,

$$u_1 = h_1(y_1, y_2), \quad \text{and} \quad u_2 = h_2(y_1, y_2), \quad (1)$$

is a one-to-one transformation from (y_1, y_2) to (u_1, u_2) with inverse

$$y_1 = h_1^{-1}(u_1, u_2), \quad \text{and} \quad y_2 = h_2^{-1}(u_1, u_2). \quad (2)$$

Multivariable Transformations Using Jacobians

The Bivariate Transformation Method

If $h_1^{-1}(u_1, u_2)$ and $h_2^{-1}(u_1, u_2)$ have continuous partial derivatives with respect to u_1 and u_2 and Jacobian

$$J = \det \begin{bmatrix} \frac{\partial h_1^{-1}}{\partial u_1} & \frac{\partial h_1^{-1}}{\partial u_2} \\ \frac{\partial h_2^{-1}}{\partial u_1} & \frac{\partial h_2^{-1}}{\partial u_2} \end{bmatrix} = \left(\frac{\partial h_1^{-1}}{\partial u_1} \right) \left(\frac{\partial h_2^{-1}}{\partial u_2} \right) - \left(\frac{\partial h_2^{-1}}{\partial u_1} \right) \left(\frac{\partial h_1^{-1}}{\partial u_2} \right) \neq 0,$$

Then the joint density of U_1 and U_2 is

$$f_{U_1, U_2}(u_1, u_2) = f_{Y_1, Y_2}(h_1^{-1}(u_1, u_2), h_2^{-1}(u_1, u_2)) |J|,$$

where $|J|$ is the absolute value of the Jacobian.

Example: Multivariable Transformations Using Jacobians

Example: Consider random variables Y_1 and Y_2 , with joint density function

$$f(y_1, y_2) = \begin{cases} 2(1 - y_1), & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $U = Y_1 Y_2$, find the probability density function for U by using the bivariate transformation method.